

Engr200 Fall 2025 Tentative Syllabus

- **Lectures:**
 - Section 1: Tuesday/Thursday 8:30-9:40, ENG Z50.
 - Section 2: Tuesday/Thursday 11:30-12:40, STD 201.

Instructor: Sinem Çöleri, scoleri@ku.edu.tr
Office hours: Tuesday/Thursday 10:00-11:00 by appointment. Please send an email to get an appointment.
 - **Problem Sessions:**
 - PSA: Tuesday 17:30-18:40, STD 201.
 - PSB: Thursday 17:30-18:40, SNA A52.

Teaching assistants: TBA
 - **Pre-requisites:** MATH 203 or consent of the instructor.
 - **Course website:** learn.hub.ku.edu.tr
The webpage includes information about the course, instructors, TAs, lecture notes, video recordings, announcements, problem assignment.
 - **Required textbook:** Introduction to Probability by D. P. Bertsekas and J. N. Tsitsiklis. Massachusetts: Athena Scientific. 2002.
 - MIT website for the course based on the textbook: <https://ocw.mit.edu/resources/res-6-012-introduction-to-probability-spring-2018>
 - **Grading:**
 - Midterm: 40%
 - Final: 50%
 - Participation: 10%
 - **Homework and PS:** Homework questions and solution videos will be posted in Learn Hub every week.
 - **Make-up exam policy:** A make-up exam will be given during final exam period to students who cannot take the exam due to an excuse deemed legitimate by the instructor. This exam will be cumulative, so that its questions can be from any of the topics covered during the semester.
 - **Academic honesty:** Honesty and trust are important to all of us as individuals. Students and faculty adhere to the following principles of academic honesty at Koc University:
 - Individual accountability for all individual work, written or oral. Copying from others or providing answers or information, written or oral, to others is cheating.
 - Providing proper acknowledgement of original author. Copying from another student's paper or from another text without written acknowledgement is plagiarism.
 - Study or project group activity is effective and authorized teamwork. Unauthorized help from another person or having someone else write one's paper or assignment is collusion.
- Cheating, plagiarism, and collusion are serious offenses resulting in an F grade and disciplinary action.
- **Participation:** Attendance will be recorded randomly throughout the semester.
 - **Tentative schedule:**

Introduction Probabilistic Models (1.1,1.2)	Cumulative Distribution Functions (3.2) Normal Random Variables (3.3)
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Conditional Probability (1.3) Total Probability Theorem and Bayes' Rule (1.4) Independence (1.5) Counting (1.6, 1.7) Probability Mass Functions (2.1,2.2) Functions of Random Variables (2.3) Mean, Variance, and Moments (2.4) Joint PMFs (2.5) Conditioning (2.6) Independence of Random Variables (2.7) Continuous Random Variables and PDFs (3.1)	Conditioning on an Event (3.4) Multiple Random Variables (3.5) Continuous Bayes' Rule (3.6) Derived Distributions (4.1) Covariance and Correlation (4.2) Conditional Expectation and Variance Revisited (4.3) Sum of a Random Number of Independent Random Variables (4.5) The Weak Law of Large Numbers (5.2) Central Limit Theorem (5.4) Random Processes Markov Chains (7.1,7.2,7.3)
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